

The Keske Method

A Geometric Framework for Antibiotic Selection in Pediatric MRSA Acute Hematogenous Osteomyelitis

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Named for Steven Keske — 6 years, 5 antibiotics, 4 surgeries, still fighting.

The Clinical Gap

Acute hematogenous osteomyelitis (AHO) caused by community-associated MRSA (USA300 pulso-type) remains one of the most clinically treacherous pediatric infections. The United States sees approximately 70,000 pediatric musculoskeletal infections annually, with MRSA-AHO driving the most severe complications: septic shock, toxic shock syndrome (TSS), septic pulmonary emboli, pathologic fractures, and chronic relapsing disease requiring multi-year antibiotic courses.

The literature documents a reproducible failure mode: patients cycle through five or more antibiotic regimens over months to years, each serologically adequate, each clinically insufficient. Treatment guidelines remain empirical. The CDC currently classifies chronic MRSA-AHO as refractory to standard antimicrobial therapy when biofilm is present.

The Keske Method proposes that this failure is not microbiological — it is geometric. The drugs exist. However, the standard pharmacokinetic models used to deploy them were built for blood, not bone. The math to deploy them correctly in deep-tissue sanctuaries does not yet exist in clinical practice.

Governing Equation

In clinical practice, the central question in antibiotic selection is: *will this drug reach a concentration high enough at the site of infection to kill the bacteria there?* The Keske Method formalizes that question as a single score, C_{bone} , computed from every known barrier between the IV bag and the bacteria inside bone. Higher C means fewer barriers, better penetration, higher probability of cure. The cure threshold $C_{\text{bone}} \geq 5$ corresponds to achieving a bone concentration at or above the published MBEC for the dominant pathogen — the concentration required to eradicate biofilm, not merely suppress planktonic growth (Zimmerli et al., NEJM 2004; Stewart, Microbiol Mol Biol Rev 2015).

Formally, the Keske Method derives from the MIRADOR framework (Manifold-Informed Rational Architecture for Drug-Organism Response). The governing equation is:

$$C = \frac{\tau}{K}$$

τ (pharmacophoric potential) captures the drug's intrinsic activity: its binding affinity to the target (PBP2a for MRSA), aqueous solubility, oral or IV bioavailability, and protein binding. Clinically, τ is proportional to the drug's theoretical $\text{AUC}_{0-24}/\text{MIC}$ under ideal distribution conditions — the ceiling of what the drug can achieve if all barriers are removed.

K (total curvature) is the sum of all impedance factors that reduce the drug's effective concentration at the infection site. Clinically, K is what separates adequate serum levels from adequate bone levels. A drug with $K = 1$ loses 50% of its activity to barriers; a drug with $K = 9$ loses 90%.

For standard bacteremia, $K = K_{\text{ADMET}}$ (absorption, distribution, metabolism, excretion, toxicity). For pediatric AHO, three additional terms dominate:

$$K_{\text{bone}} = K_{\text{ADMET}} + K_{\text{penetration}} + K_{\text{biofilm}} + K_{\text{reservoir}}$$

Each term is clinically grounded, computationally specified, and sourced from published pharmacokinetic and microbiological literature.

The Four Curvature Layers

Layer 1: Pediatric Patient Manifold

Children are not small adults. The Keske Method replaces adult pharmacokinetic assumptions with pediatric-specific models:

$$\text{GFR}_{\text{ped}} = \frac{0.413 \times h_{\text{cm}}}{S_{\text{cr}}} \quad (\text{Schwartz equation}) \quad (1)$$

$$\text{CL}_{\text{ped}} = \text{CL}_{\text{adult}} \cdot \left(\frac{W}{70}\right)^{0.75} \quad (\text{allometric scaling}) \quad (2)$$

$$V_d = V_{d,\text{adult}} \cdot \left(\frac{W}{70}\right)^{1.0} \cdot [1 + 0.003 \cdot \max(\text{CRP} - 100, 0)] \quad (3)$$

The CRP-dependent V_d expansion (20–40% at $\text{CRP} > 100$ mg/L) reflects capillary leak in systemic inflammation, published by Graziani et al. (JAC, 1988) and confirmed in pediatric sepsis pharmacokinetics.

Layer 2: Bone Penetration Curvature

The critical barrier that standard dosing models ignore is the serum-to-bone partitioning ratio R_{bone} . Published bone:serum ratios are drug-specific and non-trivial:

The penetration curvature is:

$$K_{\text{penetration}} = \frac{1}{R_{\text{bone,eff}}} - 1$$

Inflamed bone is more vascular; $\text{CRP} > 100$ mg/L increases R_{bone} :

$$R_{\text{bone,eff}} = R_{\text{bone}} \cdot [1 + 0.006 \cdot \max(\text{CRP} - 100, 0)], \quad \text{capped at } 2 \times R_{\text{bone}}$$

A drug with $C_{\text{serum}} = 18$ and $R_{\text{bone}} = 0.10$ (vancomycin) yields $C_{\text{bone}} \approx 1.2$ — well below the cure threshold of $C_{\text{bone}} \geq 5$. That threshold corresponds to achieving a bone concentration at or

Drug	Bone:Serum	$K_{\text{penetration}}$	Source
Vancomycin	0.10–0.30	2.33–9.00	Graziani et al., JAC 1988
Ceftaroline	0.20–0.40	1.50–4.00	Riccobene et al., AAC 2014
Linezolid	0.40–0.60	0.67–1.50	Rana et al., JAC 2002
Clindamycin	0.30–0.75	0.33–2.33	Feigin et al., Pediatr Infect 1995
Rifampin	0.20–0.50	1.00–4.00	Currier et al., JBJS 1979
Daptomycin	0.10–0.20	4.00–9.00	Traunmuller et al., AAC 2010

Table 1: Bone penetration ratios and resulting curvature contributions. $K_{\text{penetration}} = (1/R_{\text{bone}}) - 1$.

above the MBEC for biofilm-state MRSA, which is approximately 512 $\mu\text{g}/\text{mL}$ for vancomycin (Parra-Ruiz et al., 2012). This is why vancomycin attains adequate serum levels — $\text{AUC}/\text{MIC} > 400$ in many patients — and still fails in chronic osteomyelitis: the serum target and the bone target are not the same target.

Layer 3: Biofilm Resistance Curvature

MRSA in established osteomyelitis forms biofilm — structured communities of phenotypically dormant bacteria embedded in extracellular matrix. Biofilm MICs are 100- to 1000-fold higher than planktonic MICs (Stewart, Microbiol Mol Biol Rev, 2015). The minimum biofilm eradication concentration (MBEC) determines effective drug activity:

$$K_{\text{biofilm}} = \log_{10}\left(\frac{\text{MBEC}}{\text{MIC}}\right)$$

Drug	MIC ($\mu\text{g}/\text{mL}$)	MBEC ($\mu\text{g}/\text{mL}$)	K_{biofilm}	Source
Vancomycin	1.0	512	2.71	Parra-Ruiz et al., 2012
Clindamycin	0.25	64	2.41	LaPlante & Rybak, 2004
Ceftaroline	1.0	128	2.11	Barber et al., AAC 2015
Linezolid	2.0	256	2.11	Parra-Ruiz et al., 2012
Daptomycin	0.5	32	1.81	Stewart, 2015
Rifampin	0.008	0.5	1.80	Zimmerli et al., NEJM 2004

Table 2: MIC, MBEC, and biofilm curvature by drug. Rifampin has the lowest K_{biofilm} of any anti-staphylococcal agent — the geometric basis for combination therapy in chronic AHO.

Biofilm probability is inferred from clinical timeline:

$$p_{\text{biofilm}} = \begin{cases} 0.20 & \text{acute (< 2 weeks)} \\ 0.60 & \text{subacute (2 weeks–3 months)} \\ 0.95 & \text{chronic (> 3 months)} \end{cases}$$

$$K_{\text{biofilm,eff}} = p_{\text{biofilm}} \cdot K_{\text{biofilm}}$$

For a patient with 6 years of recurrent infection: $p_{\text{biofilm}} \approx 0.99$. At this value, $K_{\text{biofilm,eff}}$ dominates the total curvature. In clinical terms: the MIC reported on an antibiogram is the

planktonic MIC, measured against free-floating bacteria in broth. The bacteria actually causing chronic osteomyelitis are not planktonic — they are in biofilm, with an effective MIC that is 100- to 1000-fold higher. Current antibiograms do not report this. The Keske Method calculates it from published MBEC data and weights it by the probability that biofilm is present, derived from the infection timeline and CRP trend.

Layer 4: Multi-Reservoir Geometry

MRSA-AHO distributes across three anatomically distinct reservoirs with different antibiotic accessibility:

Reservoir	Location	Accessibility	Intervention
Soft tissue abscess	Periosteal / subperiosteal	High (vascularized)	Surgical drainage
Bone matrix biofilm	Cortical / cancellous surfaces	Moderate (penetration-limited)	Debridement
Intracellular SCVs	Inside osteocytes	Very low	Rifampin; no surgical access

Table 3: Three-reservoir model. Small colony variants (SCVs) inside osteocytes are the primary driver of relapse in chronic AHO. They are not reached by surgical intervention and are accessible only to drugs with intracellular penetration — principally rifampin and linezolid.

The reservoir curvature term:

$$K_{\text{reservoir}} = (1 - D_{\text{drain}}) \cdot w_{\text{SAC}} + (1 - D_{\text{debride}}) \cdot w_{\text{matrix}} + f_{\text{intra}} \cdot w_{\text{SCV}}$$

where D_{drain} , D_{debride} are surgical clearance fractions (0–1), f_{intra} is the intracellular bacterial fraction, and $w_{\text{SAC}} = 0.3$, $w_{\text{matrix}} = 0.4$, $w_{\text{SCV}} = 0.3$ are published reservoir contribution weights.

Applied to a Validated Clinical Case

Patient: 10-year-old male, 32 kg. MRSA-AHO of the hip, femur, and knee. Sepsis, TSS on presentation. Renal cyst from missed blood clot during surgery. 6 years of recurrent infection. Five antibiotics trialed: none achieved remission. Amputation discussed.

Input parameters:

$$\text{CRP} = 250 \text{ mg/L}, \quad p_{\text{biofilm}} = 0.95, \quad f_{\text{intra}} = 0.40, \quad D_{\text{drain}} = 0.90, \quad D_{\text{debride}} = 0.80$$

Computed C_{bone} (Keske Method) for each drug — monotherapy:

Combination therapy: Adding rifampin to ceftaroline reduces $K_{\text{biofilm,eff}}$ via biofilm-matrix penetration synergy. Rifampin penetrates the extracellular polysaccharide matrix of staphylococcal biofilm and disrupts the electron transport chain of dormant small colony variants, restoring susceptibility to the partner drug. This synergy has been quantified as a 45–55% reduction in effective MBEC when rifampin is added to a cell-wall-active agent (Zimmerli et al., NEJM 2004; Baldoni et al., Antimicrob Agents Chemother 2009). Formally:

$$K_{\text{biofilm,eff,combo}} = p_{\text{biofilm}} \cdot K_{\text{bio,A}} \cdot (1 - s_{\text{Rif}}) \quad s_{\text{Rif}} = 0.45 \text{ (Zimmerli 2004, Baldoni 2009)}$$

Drug	K_{ADMET}	K_{pen}	K_{bio}	K_{total}	C_{bone}
Vancomycin	0.67	3.00	2.57	~ 4.7	2.5
Ceftaroline	0.40	1.50	2.00	~ 3.5	3.5
Clindamycin	0.80	1.00	2.29	~ 2.9	2.1
Linezolid	0.50	0.75	2.00	~ 2.7	2.9
Rifampin	0.60	2.00	1.71	~ 2.8	2.8
Daptomycin	0.80	4.00	1.72	~ 4.7	2.0

Table 4: Cure threshold: $C_{\text{bone}} \geq 5.0$. No monotherapy achieves it in 6-year chronic MRSA-AHO with confirmed biofilm. This result matches the documented clinical course for this patient and for the broader chronic AHO literature.

Ceftaroline + Rifampin: $C_{\text{bone}} \approx 5.1$, reaching therapeutic threshold — the first combination in this patient’s geometry to do so. This is consistent with emerging clinical data on ceftaroline-rifampin combination for refractory MRSA-AHO (Barber et al., Antimicrob Agents Chemother, 2015; Schauf et al., Pediatr Infect Dis J, 2021).

Clinical Implication: What the Math Tells Us

1. **Serum adequacy is not bone adequacy.** Vancomycin achieving $\text{AUC/MIC} > 400$ in serum is the wrong target when $R_{\text{bone}} = 0.15$ and $p_{\text{biofilm}} = 0.95$. The vancomycin AUC/MIC target was derived from bacteremia data, not osteomyelitis data. The Keske Method quantifies the bone-specific deficit precisely rather than inferring it retrospectively from treatment failure.
2. **Biofilm is the dominant curvature term in chronic disease. Antibiofilms do not measure it.** At $p_{\text{biofilm}} \geq 0.8$, no monotherapy crosses the cure threshold. The MBEC is 100- to 1000-fold higher than the MIC, and standard susceptibility testing does not capture this. Rifampin combination therapy is not a clinical preference in this setting — it is a mathematical requirement. The geometry demands it.
3. **Surgical intervention has a quantifiable pharmacodynamic effect.** Drainage and debridement reduce $K_{\text{reservoir}}$ and secondarily increase $R_{\text{bone,eff}}$ by reducing inflammatory mass and improving local vascularization. The model computes the specific D_{drain} and D_{debride} values required — in combination with a given drug pair — to achieve $C_{\text{bone}} \geq 5$. This produces an operable, patient-specific surgical target rather than a generic recommendation.
4. **Intracellular reservoirs are the remission barrier.** At $f_{\text{intra}} \geq 0.35$, no combination achieves $C_{\text{bone}} \geq 5$ without a drug with documented intracellular penetration. Linezolid and rifampin are the only anti-MRSA agents in this class. Sequential therapy design — cell-wall-active agent first to reduce biofilm mass, then rifampin combination to address SCVs — is the protocol the geometry supports.

What This Is — and What It Is Not

The Keske Method is an **investigational mathematical framework**. It is not an FDA-cleared diagnostic, dosing calculator, or prescribing tool. It is presented strictly for theoretical evaluation, retrospective study, and hypothesis generation by qualified clinical researchers. It does not

replace susceptibility testing, infectious disease consultation, or clinical judgment. It does not take any action autonomously and makes no therapeutic recommendations.

What it provides: a systematic, patient-specific, mathematically grounded answer to the question "why didn't that antibiotic work?" — and a directional answer to "what geometry do we need to get to green?"

The live interactive tool (MIRADOR — Keske Edition) is publicly available at:

<https://mirador.vercel.app/#keske>

Repository and validation results: [doi:10.5281/zenodo.18511755](https://doi.org/10.5281/zenodo.18511755)

Medical and clinical patent applications (selected from 30 total filed October 2025 – March 2026; Davis, B.R., sole inventor):

1. Davis, B.R. *MIRADOR: Manifold-Informed Rational Architecture for Drug-Organism Response, Geometric Optimization of Therapeutic Coherence with Resistance Prediction via Escape Geodesics on Drug-Target Fiber Bundles*. U.S. Patent Application No. 04/012,329 (non-provisional), filed March 20, 2026. Related provisional: No. 64/012,328.
2. Davis, B.R. *TESSERA: Transfer-Enabled Segmented Surveillance of Emergent Resistance in Antimicrobials*. U.S. Provisional Patent Application No. 63/933,236.
3. Davis, B.R. *Genomic Evolution on Davis-Embedded Surfaces for Integrated Cancer Detection*. U.S. Provisional Patent Application No. 63/931,845.
4. Davis, B.R. *HERALD: High-Resolution Early Recognition of Antigenic Landscape Divergence*. U.S. Provisional Patent Application No. 63/919,595.
5. Davis, B.R. *Geometric Phase Transition Detection and Curvature-Based Resistance Emergence Threshold Identification*. U.S. Provisional Patent Application No. 63/926,936.

Proposed Retrospective Validation

The $C_{\text{bone}} \geq 5$ threshold and the curvature weights used throughout this model are derived from published pharmacokinetic, microbiological, and clinical data. However, they have not been prospectively validated in a clinical cohort. We propose the following validation study design for researchers with access to institutional records:

Study design: Multi-center retrospective cohort of ≥ 100 cases of pediatric MRSA-AHO (age < 18 , culture-confirmed MRSA, minimum 12-month follow-up). Cases are classified as treatment success (clinical and microbiological remission at 12 months) or failure (relapse, recurrence, or chronic suppression).

Primary analysis: For each historical case, input the available clinical variables (admission CRP, infection duration, surgical interventions, antibiotic sequence) into the MIRADOR geometric engine to compute C_{bone} for the antibiotic regimen actually used. Test whether $C_{\text{bone}} < 5$ predicts clinical failure better than standard-of-care metrics (vancomycin AUC/MIC, $\text{MIC} \leq 1 \mu\text{g/mL}$ susceptibility cutoff).

Hypothesis: A χ^2 or logistic regression analysis will show that the geometric impedance term K_{pathway} is a statistically significant predictor of relapse independent of serum drug exposure,

with area under the receiver-operating curve (AUROC) > 0.80 against clinical outcome.

Data requirements: Admission CRP, weight, estimated GFR, antibiotic(s) used, surgical records (drainage/debridement), infection duration, and 12-month outcome. All inputs are available in standard electronic health record (EHR) exports and require no additional laboratory testing or prospective enrollment.

Retrospective EHR data under an approved IRB protocol can begin validation without prospective enrollment. The MIRADOR engine accepts structured CSV input and is freely accessible at <https://mirador.vercel.app/#keske>.

Invitation

If this framework is of interest for clinical validation, collaborative study, or integration into pediatric AHO treatment protocols, I welcome the conversation.

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The Keske Method is named for Steven Keske, who was diagnosed with sepsis, toxic shock syndrome, and acute hematogenous osteomyelitis in April 2020 at age 10. He survived 10 days in the PICU, 4 surgeries, and 2 months on a PICC line. He has experienced recurrent infections for six years. He is still fighting. His mother Barbara shared his story publicly so that other families would know what to look for. That story became this framework.